

Interactive Learning Platform for Ethiopian Diaspora Children

A voice-first, gamified web experience grounded in verified cultural heritage and adaptive AI tutoring.

TARGET AUDIENCE Ethiopian Diaspora Youth (Ages 6–14)	CORE MODALITY Voice-First + Multimodal Visuals	PRIMARY LANGUAGE English & Amharic (MVP)
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Final Product Vision: A voice-first interactive gamified web learning experience where Ethiopian diaspora children discover history, heritage, and nature through short stories, visuals, questions, and quizzes, with AI responses strictly grounded in verified sources.

Research-Driven Product Decisions (Part 1)

1. Interactive Learning & Cultural Heritage ScholarXIV Papers Key Finding: Research suggests interactive storytelling can make cultural heritage learning more engaging than static presentation. Product Decision: Present Ethiopian history via short, interactive game stories. High-complexity technologies like AR/VR are omitted from the MVP to deliver a polished web prototype.
2. Voice Interaction for Children Voice-Controlled Scratch & ChildBot Key Finding: Voice increases accessibility, but speech recognition can struggle with young learners without structured bounds and visual feedback support. Product Decision: Voice (powered by Voxide) is a core interaction. Kids ask questions naturally using voice, supported by synchronized text, graphics, and audio cues.
3. AI Accuracy & Grounded Answers GenAI Voice Mode & Verifiability Research Key Finding: Generative AI can produce factually incorrect information, making source-grounded responses important for educational content. Product Decision: The AI tutor will not answer from ungrounded memory. All responses retrieve verified data from a curated Ethiopian knowledge base with verified sources and source-level tracking.

Research-Driven Product Decisions (Part 2)

4. Child-Friendly & Adaptive Scaffolding	SHAPE Educational LLMs
Key Finding: Effective AI pedagogy guides students through hint-based scaffolding rather than giving direct answers immediately.	
Product Decision: Game loops integrate progressive hinting, follow-up dialogue, adaptive difficulty, and short check-for-understanding quizzes.	
5. Multilingual Voice Learning	CodeVaani Multilingual Assistant
Key Finding: Multilingual audio bridging reduces language barriers for learners navigating non-native primary languages.	
Product Decision: Launch with English and Amharic voice support tailored for diaspora children, with scalable architecture for adding Afaan Oromo, Tigrinya, and other languages.	

Synthesis Matrix: Research → Product Logic

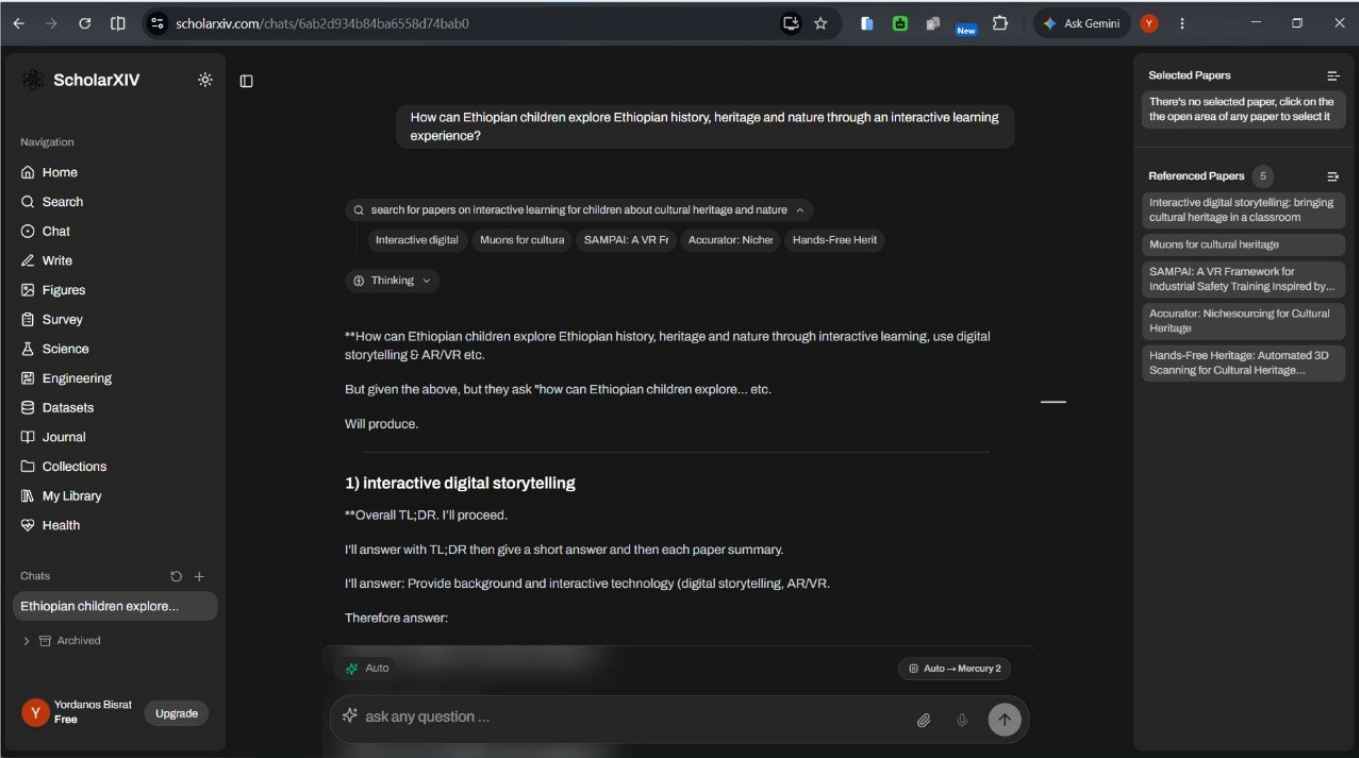
Research Findings	Gamified Product Implementation
Interactive storytelling improves engagement vs. static reading.	Interactive Narrative Map: Quest-based exploration of Lalibela, Axum, and Ethiopian wildlife.
Voice reduces friction for children when paired with visual context.	Voice-First Companion: Kids talk directly to an animated AI guide using Voxide.
Generative AI requires factual grounding to prevent misinformation.	RAG Knowledge Base: Curated Ethiopian knowledge base with verified sources (e.g., UNESCO, NALA, Ethiopian academic publications, EBI, Wikidata) and source-level tracking.
Pedagogical scaffolding promotes long-term retention.	Adaptive Gamification: Earn badges through voice quizzes, hints, and interactive dialogue.
Multilingual speech engines bridge cultural disconnects for diaspora youth.	Dual-Language Mode: Toggle seamlessly between English explanations and Amharic vocabulary.

Appendix: ScholarXIV Research Evidence

Screenshots of the six ScholarXIV queries behind the product decisions above.

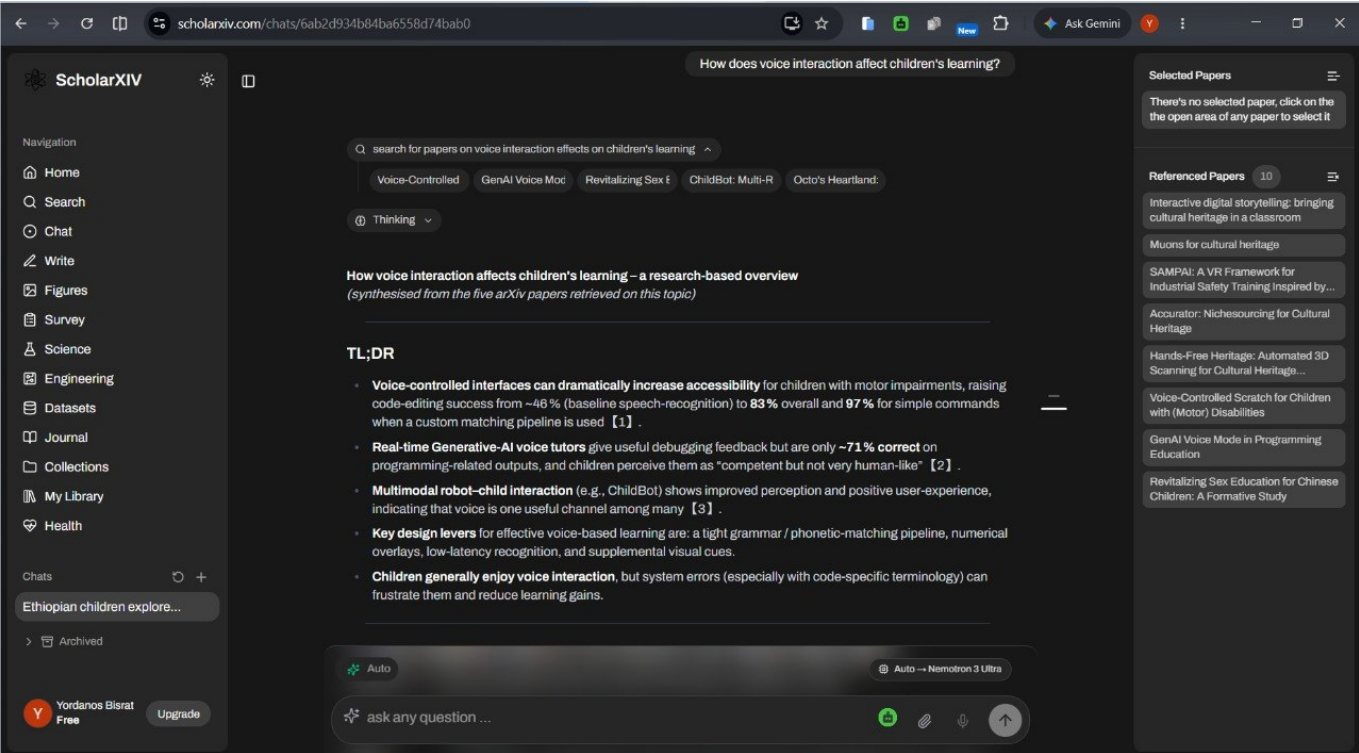
Query 1 · Interactive Learning & Cultural Heritage

Supports Decision 1: interactive digital storytelling for heritage, with AR/VR deferred past the MVP.



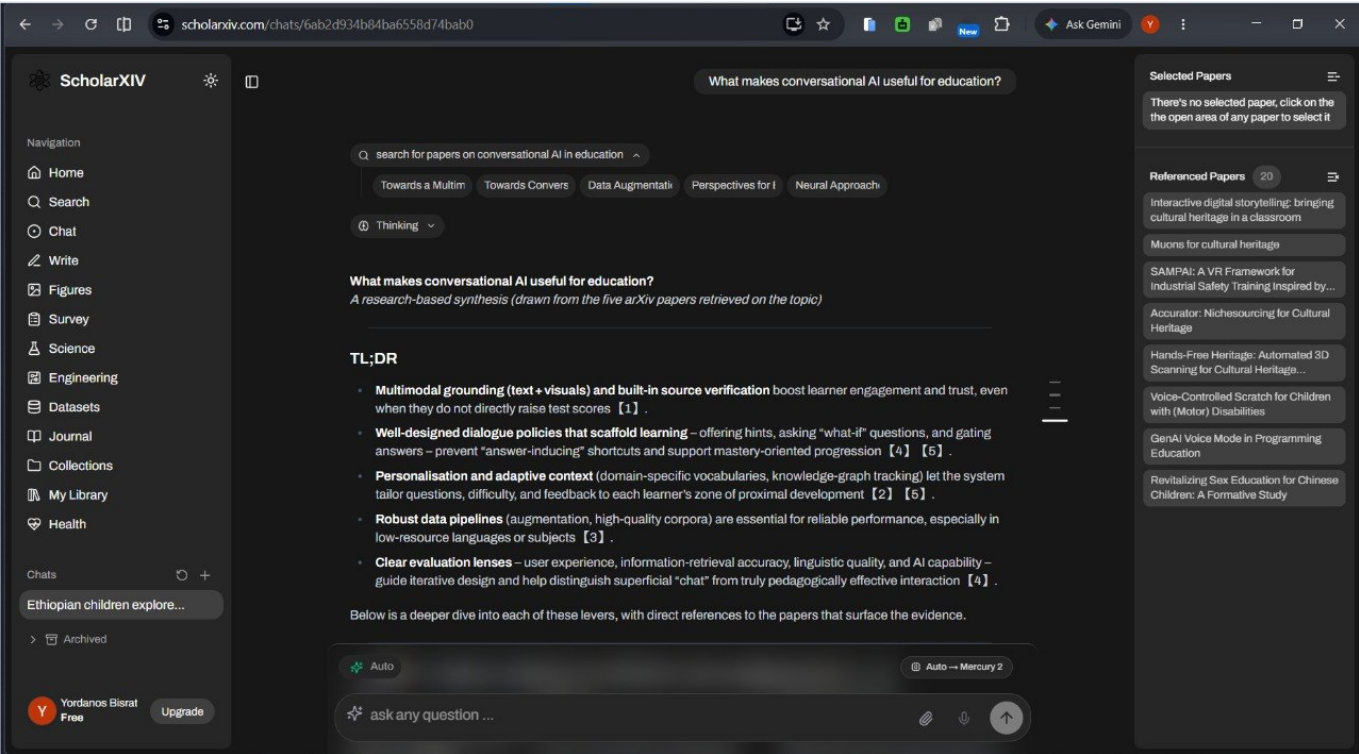
Query 2 · Voice Interaction & Children's Learning

Supports Decision 2: voice-controlled interfaces (Voice-Controlled Scratch, ChildBot) and the need for visual feedback.



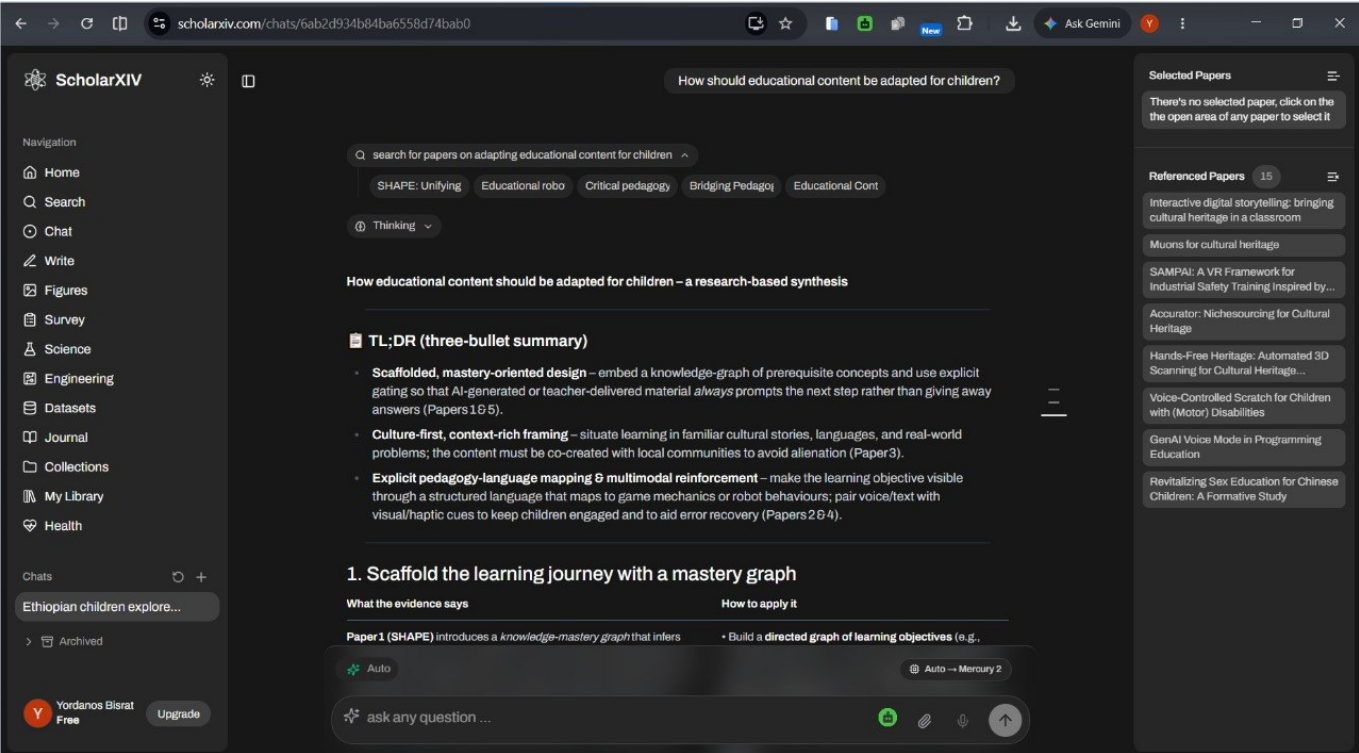
Query 3 · Conversational AI in Education

Supports Decision 3: multimodal grounding and built-in source verification for trustworthy AI answers.



Query 4 · Adapting Educational Content for Children

Supports Decision 4: SHAPE-style mastery graphs, hint-based scaffolding and culture-first framing.



Query 5 · Multilingual Voice Interaction

Supports Decision 5: CodeVaani and multilingual voice support for learners outside their primary language.

ScholarXIV

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How can multilingual voice interaction support learning?

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CodeVaani: A Mu

GenAI Voice Mod

Contrastive Learn

Common Voice: A

FreeSVC: Toward

Thinking

How multilingual voice interaction can support learning – a research-based overview

TL;DR

Multilingual voice interfaces widen access to educational content by letting learners interact in their native tongues, increase engagement through affective and contextual cues, and—when paired with robust speech-recognition pipelines and curated multimodal resources—support mastery-oriented progression in domains such as programming, language acquisition, and broader STEM learning. The five recent arXiv studies illustrate concrete designs (code-focused assistants, real-time GenAI tutors, face-voice linking, large speech corpora, and zero-shot voice conversion) that together show how language-diverse voice interaction can be leveraged for teaching and learning.

1. CodeVaani – a multilingual voice-based code-learning assistant

arXiv 2511.20654v1 (2025) – CodeVaani: A Multilingual, Voice-Based Code Learning Assistant

Goal: Enable programming education for students who are not proficient in English, focusing on the Indian multilingual context.

Auto

Auto → Mercury 2

ask any question ...

Selected Papers

There's no selected paper, click on the the open area of any paper to select it

Referenced Papers 24

Interactive digital storytelling: bringing cultural heritage in a classroom

Muons for cultural heritage

SAMPAT: A VR Framework for Industrial Safety Training Inspired by...

Accurator: Nichesourcing for Cultural Heritage

Hands-Free Heritage: Automated 3D Scanning for Cultural Heritage...

Voice-Controlled Scratch for Children with (Motor) Disabilities

GenAI Voice Mode in Programming Education

Revitalizing Sex Education for Chinese Children: A Formative Study

Query 6 · Gamification & Children's Engagement

Supports the Adaptive Gamification row: badges, leaderboards and quizzes, with design caveats on over-reliance.

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search for papers on gamification effects on children's engagement

Quantum Port: G

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How gamification affects children's engagement – a research-based overview

(synthesised from the five arXiv papers retrieved on the topic)

TL;DR (3-bullet summary)

- Well-designed gamification boosts motivation, participation and time-on-task – points, badges, leaderboards, and clear narrative contexts are the most frequently cited levers across the studies.
- Design matters more than the mere presence of game elements – competition can raise engagement for some children but may demotivate others; cooperation and scaffolded challenges tend to produce more consistent positive effects.
- Poorly implemented gamification can backfire – over-reliance on leaderboards, premature reward of completion, or mis-aligned mechanics may reduce intrinsic motivation and even lower performance (the “overjustification” effect).

1. Paper-by-paper summaries

Core focus 6

Key gamification

Auto

Auto → Nemotron 3 Ultra

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